Symulacja i analiza rozkładów cząstek pp @ $\sqrt{s} = 13.6$ TeV

Konstanty Sobczyński

Ustawienia symulacji

- Pythia 8.312, ROOT 6.40.02
- zderzenia pp, `Beams:eCM = 13600` GeV
- *minimum bias*: `SoftQCD:inelastic = on`
- 100 000 zderzeń, seed 123456
- analizowane wyłącznie cząstki stanu końcowego (`isFinal()`)

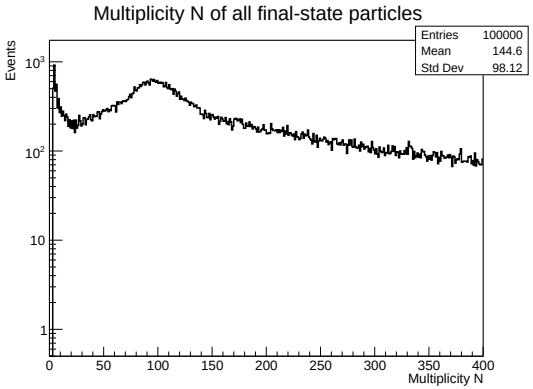
Łańcuch analizy

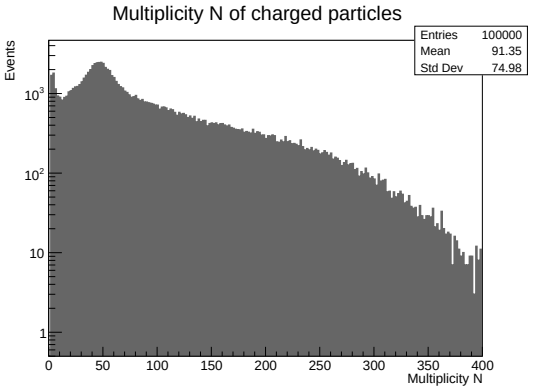
`generate` → `events.root` → `analysis` → `analysis.root` → `draw_plots.C` → wykresy

kategoria	wybór
all	wszystkie stanu końcowego
ch	ładunek $\neq 0$
$\pi^{\pm}$	PDG ± 211
$K^{\pm}$	PDG ± 321
$p,\ \bar{p}$	PDG ± 2212

wielkość	zakres	biny
N	0–400	400
p_T	0–10 GeV	100
η	-5–5	100
ϕ	0–π	64

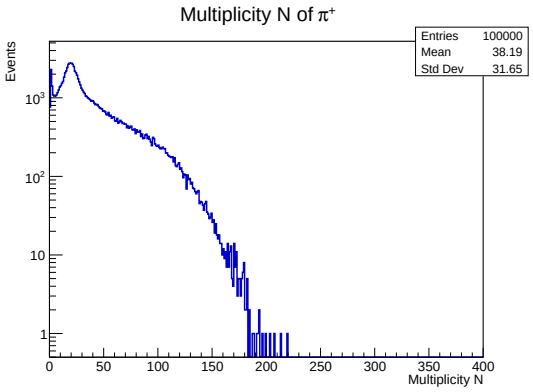
Krotność N: wszystkie i naładowane

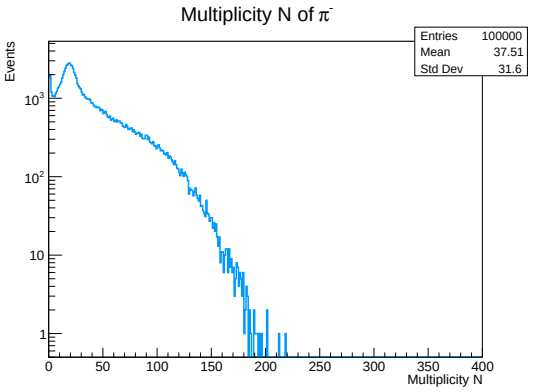




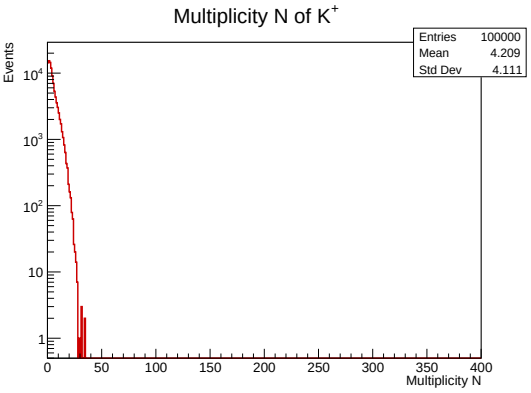
12.5% zderzeń ma $N_{\mathrm{all}} \geq 400$ (poza zakresem); *Mean* w ramce nie uwzględnia tych zderzeń ($\langle N_{\mathrm{all}} \rangle \approx 192$). N_{ch} jest zawsze parzyste, stąd co drugi bin pusty.

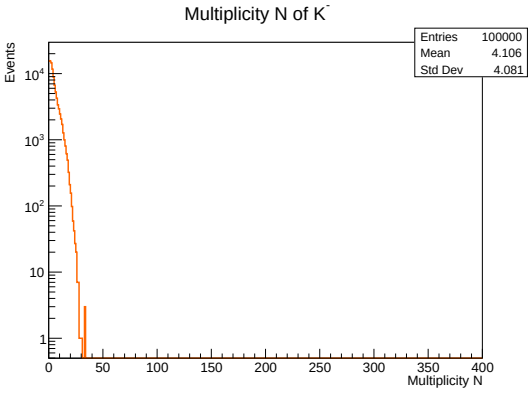
Krotność N: piony



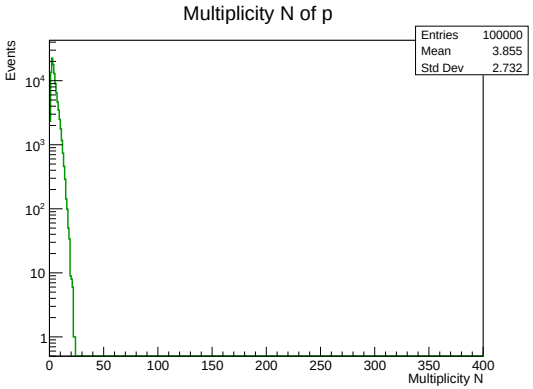


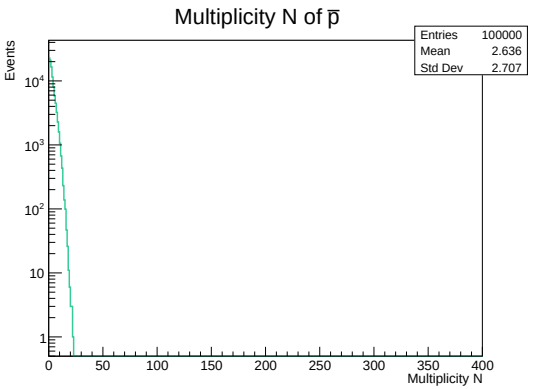
Krotność N: kaony



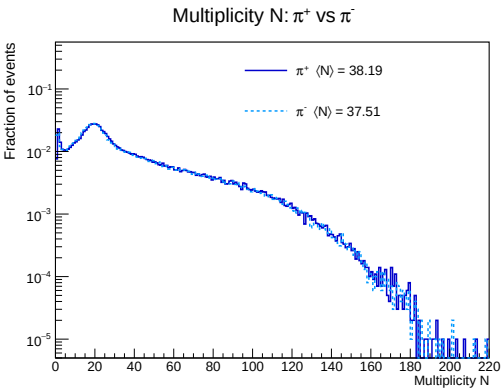


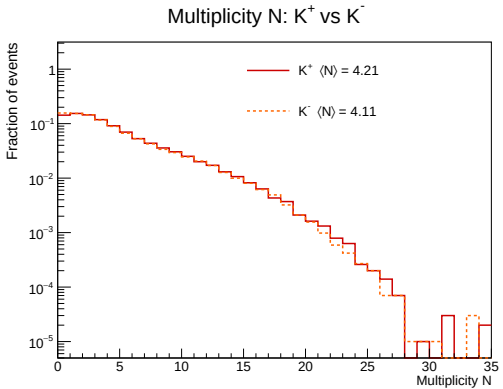
Krotność N: protony





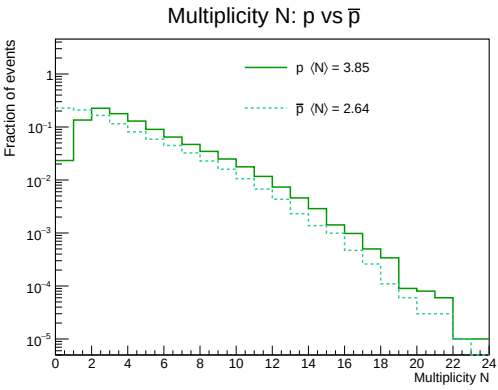
Krotność N: porównania

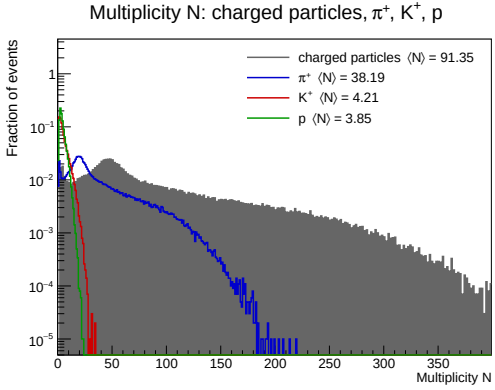




Histogramy podzielone przez liczbę zderzeń.

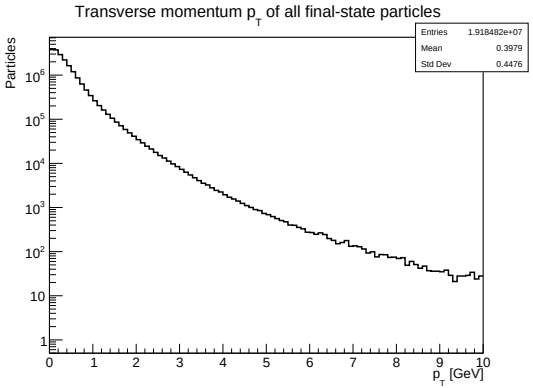
Krotność N: porównania

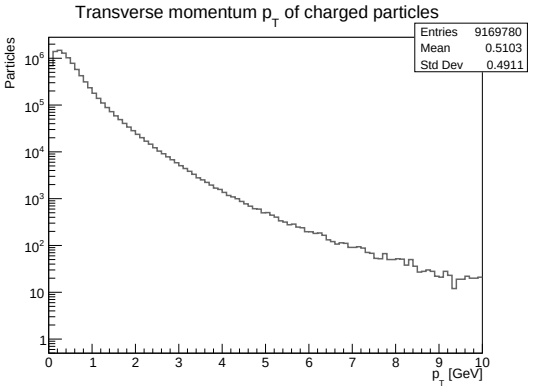




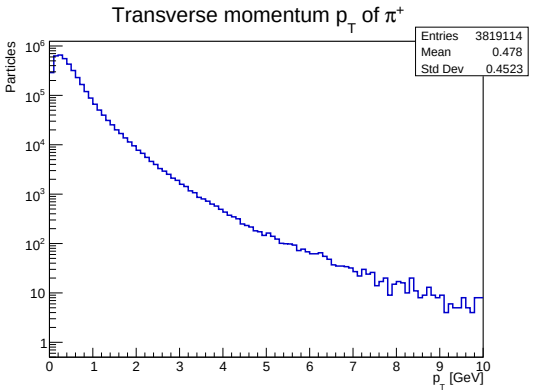
Histogramy podzielone przez liczbę zderzeń.

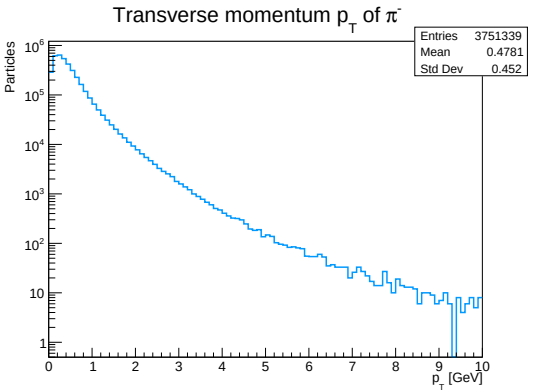
Pęd poprzeczny p_T: wszystkie i naładowane



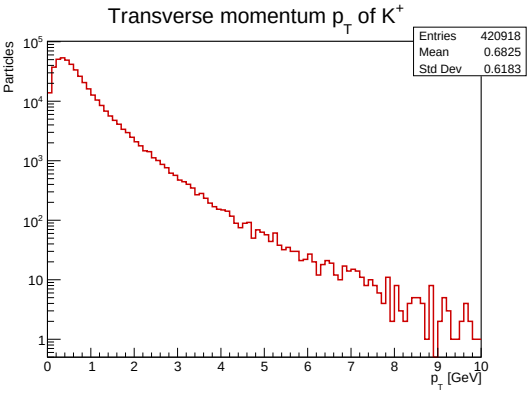


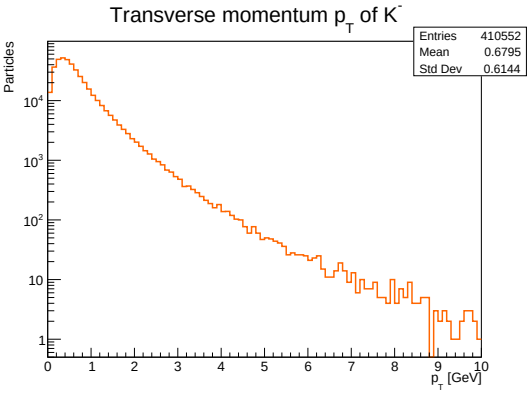
Pęd poprzeczny p_T: piony



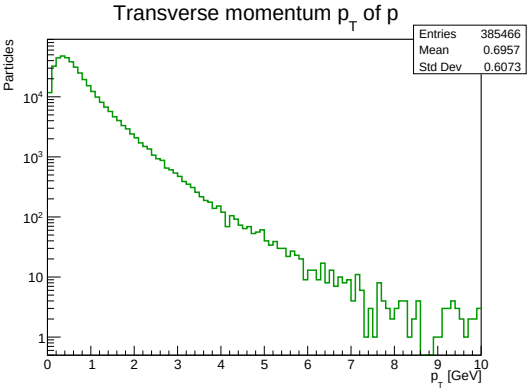


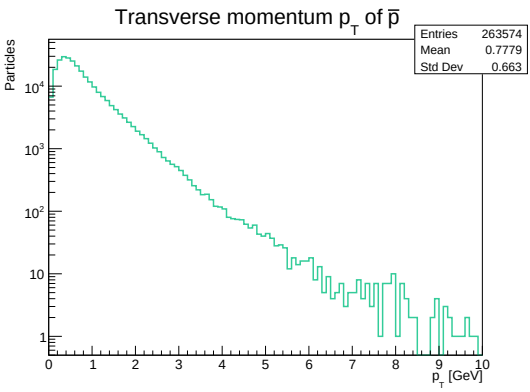
Pęd poprzeczny p_T: kaony



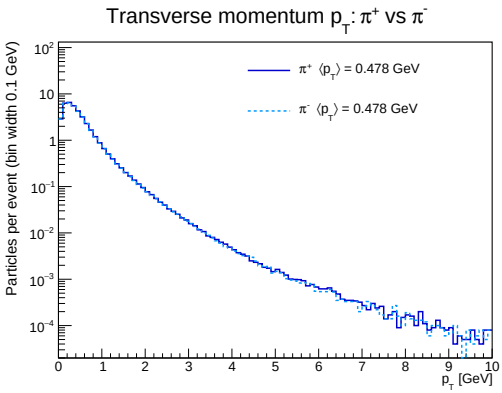


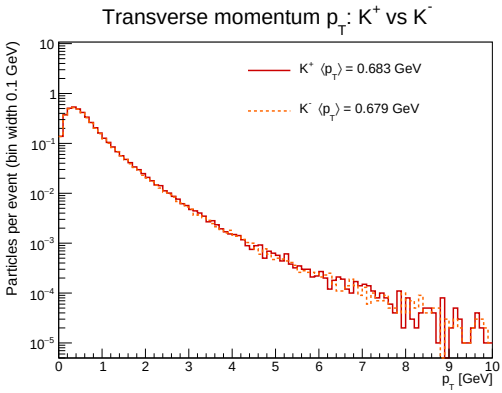
Pęd poprzeczny p_T: protony





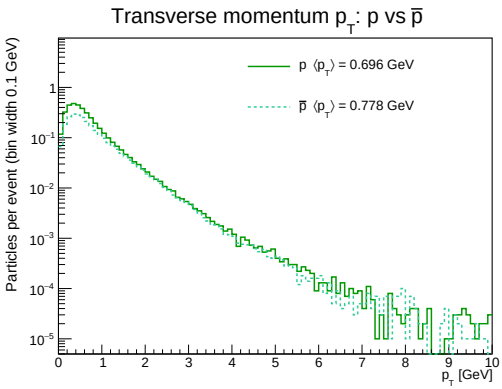
Pęd poprzeczny p_T: porównania

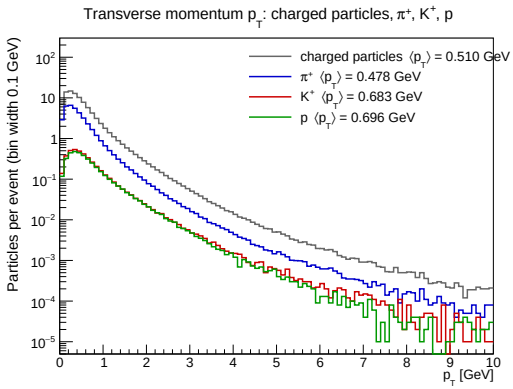




Histogramy podzielone przez liczbę zderzeń.

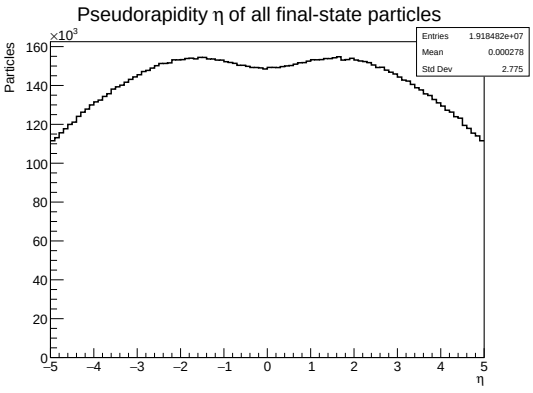
Pęd poprzeczny p_T: porównania

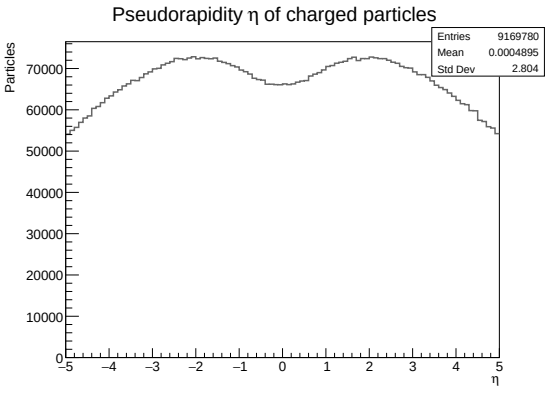




Histogramy podzielone przez liczbę zderzeń.

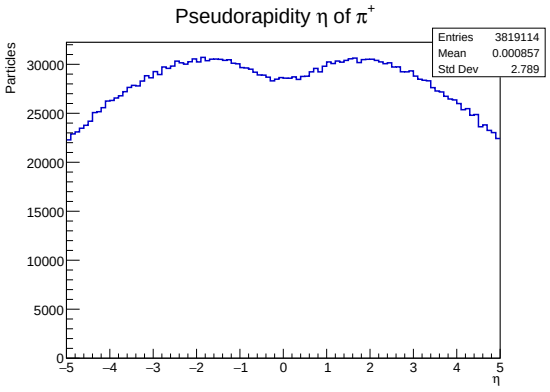
Pseudorapidity η: wszystkie i naładowane

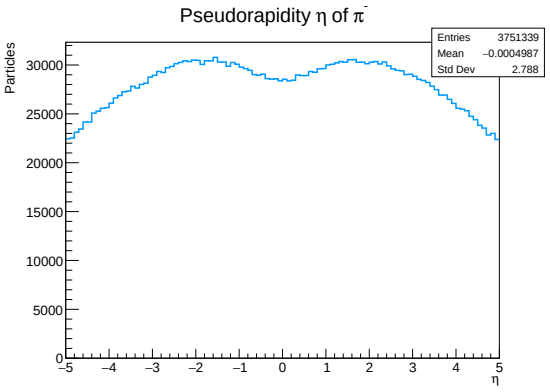




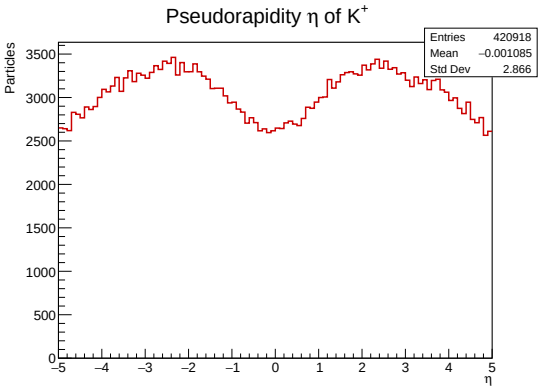
Ok. 26% cząstek ma $|\eta| > 5$ (poza zakresem).

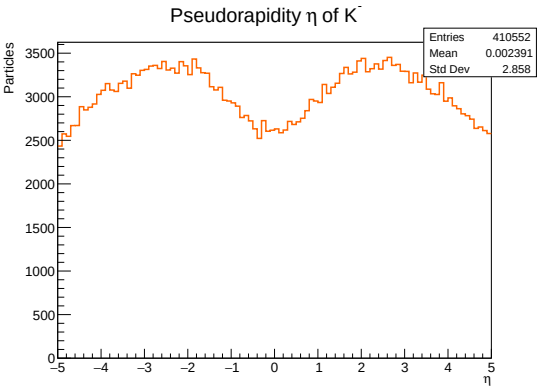
Pseudorapidity η: piony



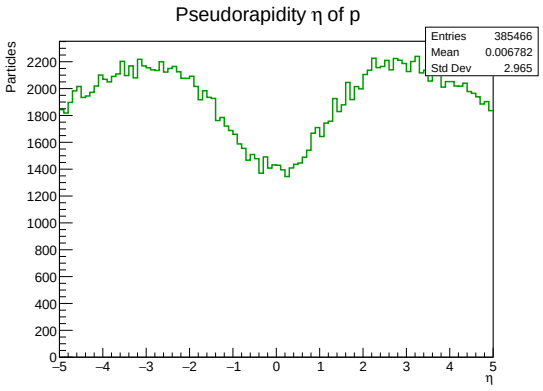


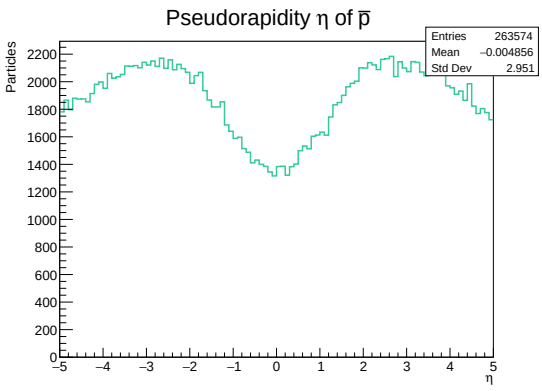
Pseudorapidity η: kaony



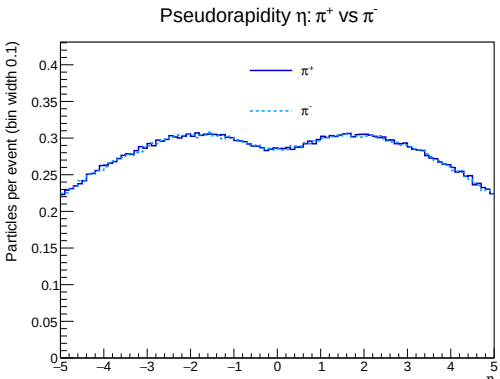


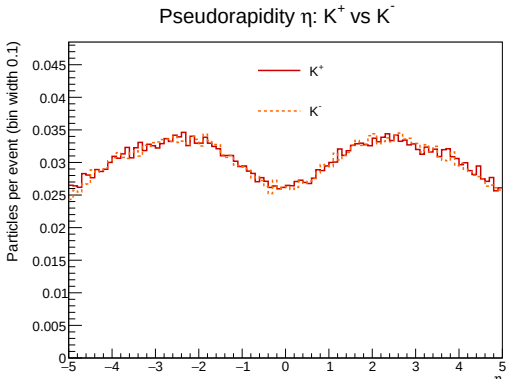
Pseudorapidity η: protony





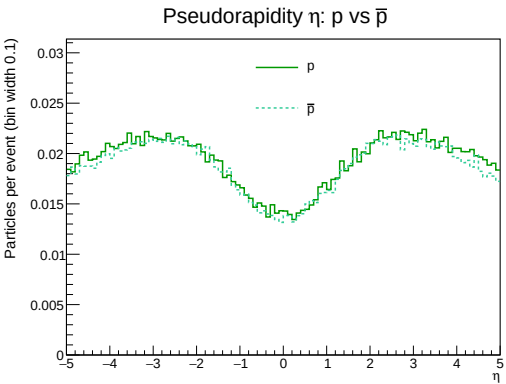
Pseudorapidity η: porównania

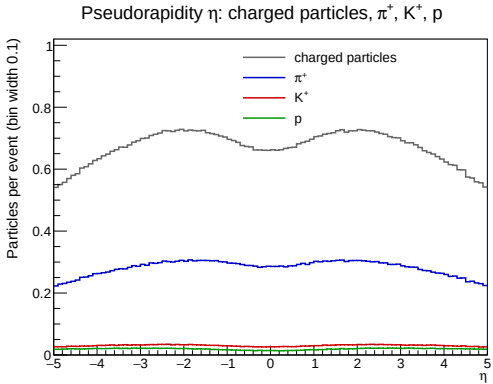




Histogramy podzielone przez liczbę zderzeń.

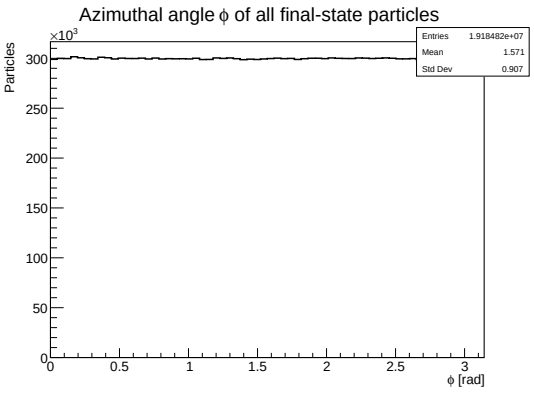
Pseudorapidity η: porównania

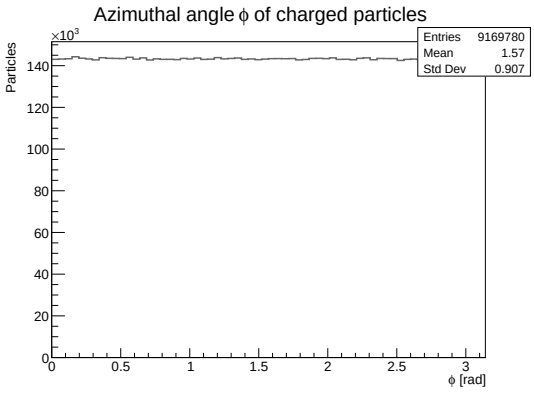




Histogramy podzielone przez liczbę zderzeń.

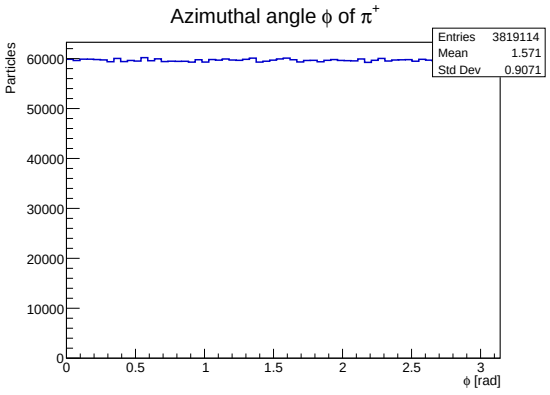
Kąt azymutalny ϕ: wszystkie i naładowane

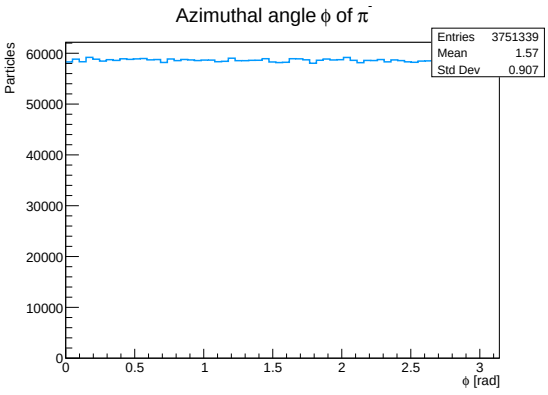




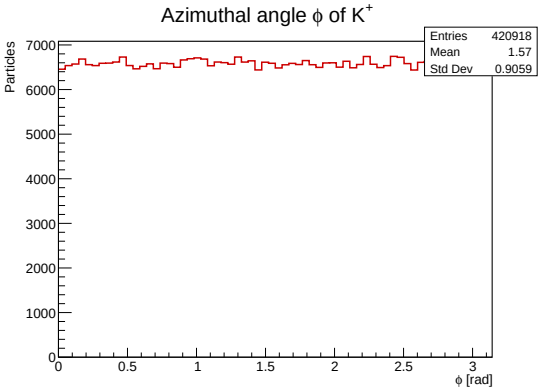
ϕ złożone do zakresu $[0, \pi]$ przez $|\phi|$.

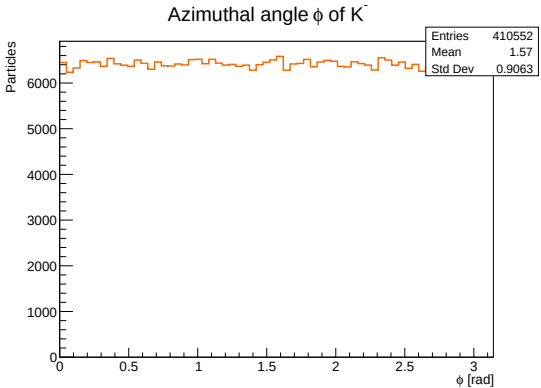
Kąt azymutalny ϕ: piony



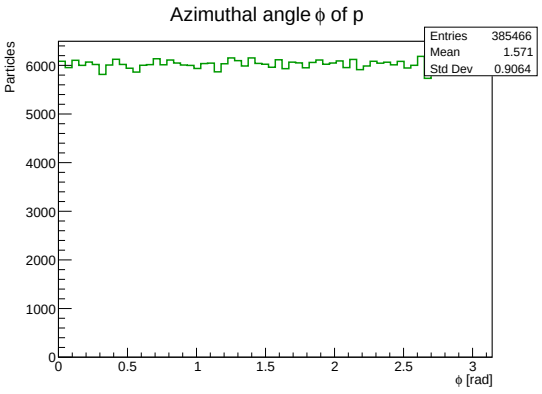


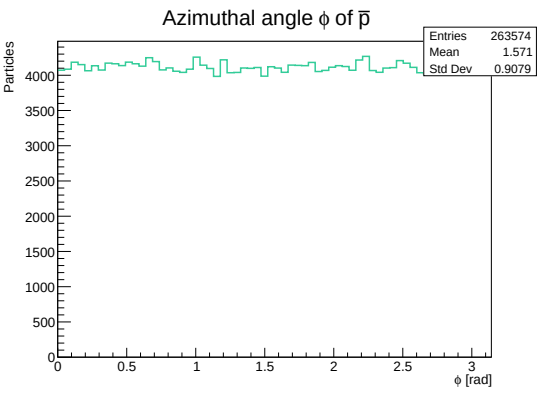
Kąt azymutalny ϕ: kaony



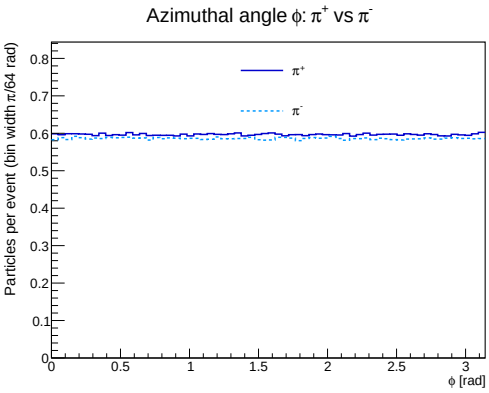


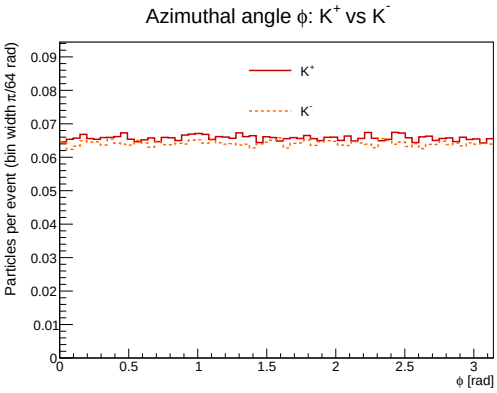
Kąt azymutalny ϕ: protony





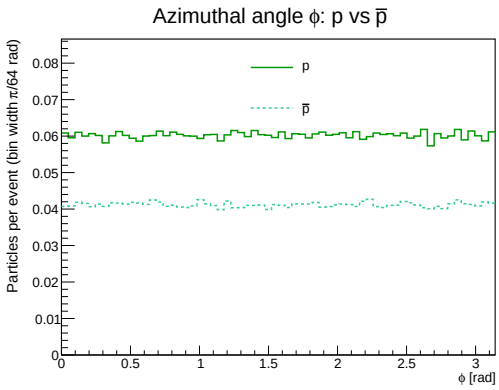
Kąt azymutalny ϕ: porównania

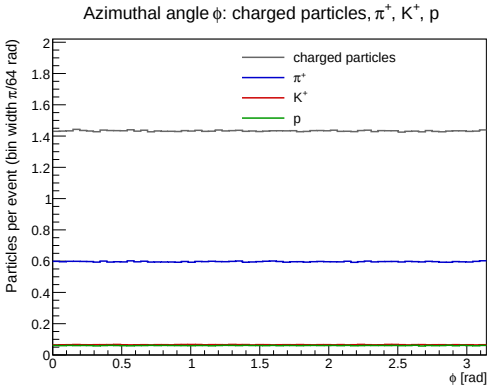




Histogramy podzielone przez liczbę zderzeń.

Kąt azymutalny ϕ: porównania





Histogramy podzielone przez liczbę zderzeń.

Podsumowanie liczbowe

kategoria	$\langle N \rangle$ na zderzenie	$\langle p_T \rangle$ [GeV]
wszystkie stanu końcowego	191.9	0.398
naładowane	91.7	0.510
π^+ / π^-	38.19 / 37.51	0.478 / 0.478
K^+ / K^-	4.21 / 4.11	0.683 / 0.679
p / $\bar{p}$	3.85 / 2.64	0.696 / 0.778

$\langle N \rangle$ liczone ze wszystkich zderzeń, łącznie z tymi poza zakresem histogramu.